

FORM TP 9288

JUNE 1992

CARIBBEAN EXAMINATIONS COUNCIL

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

MATHEMATICS

Paper 2 - General Proficiency

2 hours 40 minutes

1. Answer ALL the questions in Section I, and any TWO in Section II.
2. Begin the answer for each question on a new page.
3. Full marks may not be awarded unless full working or explanation is shown with the answer.
4. Mathematical tables, formulae and graph paper are provided.
5. Slide rules and silent electronic calculators may be used for this paper.
6. You are advised to use the first 10 minutes of the examination time to read through this paper. Writing may begin during this 10-minute period.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

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SECTION I

Answer ALL the questions in this section.

1. All working must be clearly shown.

(a) Calculate the exact value of

$$\frac{5\frac{2}{7} + 3\frac{5}{7}}{4 - 2\frac{4}{5}} \quad (4 \text{ marks})$$

(b) A piece of string 64 cm long, is divided in three pieces in the ratio 1 : 2 : 5. Calculate the length of the longest piece. (3 marks)

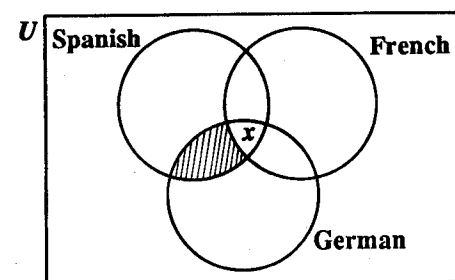
(c) A merchant sold a pen for \$5.35, thereby making a profit of 7% on the cost to him. Calculate

- (i) the cost price of the pen to the merchant
- (ii) the selling price the merchant should request in order to make a 15% profit. (5 marks)

2. (a) Solve $\frac{2x-1}{2} - \frac{x+5}{3} = 2$. (4 marks)

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(b)



The Venn diagram above illustrates some of the information given below.

There are 100 members in a foreign language club.

- 48 members speak Spanish.
- 45 members speak French.
- 52 members speak German.
- 15 members speak Spanish and French.
- 18 members speak Spanish and German.
- 21 members speak German and French.
- Each member speaks AT LEAST ONE of the three languages.

Let the number of members who speak all three languages be x .

- (i) Write an algebraic expression to represent the number of members in the shaded region.
- (ii) Describe the region shaded.
- (iii) Write an equation to show the total number of members in the club.
- (iv) Hence, determine the number of members who speak all three languages. (7 marks)

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3. (a) Solve the simultaneous equations:

$$\begin{aligned} 4x - 4y &= 2 \\ 7x + 2y &= 17 \end{aligned}$$

(4 marks)

- (b) Factorize completely:

(i) $1 - 9x^2$

(ii) $3x^2 - 7x - 6$

(3 marks)

- (c) The vector $\underline{p}$ translates the point (2, 3) to the point (4, 6).

The vector $\underline{q}$ translates the point (2, 7) to the point (-4, 3).

- (i) Write $\underline{p}$ and $\underline{q}$ as column vectors.

- (ii) Hence, determine the vector $\frac{1}{2}(\underline{p} + \underline{q})$. (5 marks)

4. (a) (i) Using ruler and compasses only, construct a quadrilateral $ABCD$ in which $AB = AD = 6$ cm, $BC = 4$ cm, angle $BAD = 60^\circ$ and angle $ABC = 90^\circ$.

- (ii) Measure and state

– the length of DC

– the size of angle ADC .

(6 marks)

- (b) The images of $L(1, 1)$ and $N(2, 3)$ under a single transformation Q are $L'(-1, 1)$ and $N'(-3, 2)$ respectively.

- (i) Describe geometrically the transformation Q .

- (ii) Determine the equation of the line $L'N'$. (6 marks)

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5. (a) Copy in your answer booklet the table below for $f(x) = 2x^2 - x - 3$ for the domain $-2 \leq x \leq 3$, and calculate the missing values.

x	-2	-1	0	$\frac{1}{2}$	1	2	3
$f(x)$	7				-2	3	12

- (b) Using a scale of 2 cm to represent 1 unit on the x -axis and 1 cm to represent 1 unit on the $f(x)$ -axis, draw the graph of $f(x)$ for $-2 \leq x \leq 3$.

- (c) From your graph, determine

- (i) the values of x for which $f(x) = 0$

- (ii) the minimum value of $f(x)$. (11 marks)

Note: You should attach your graph carefully to the page on which you did the other working for this question.

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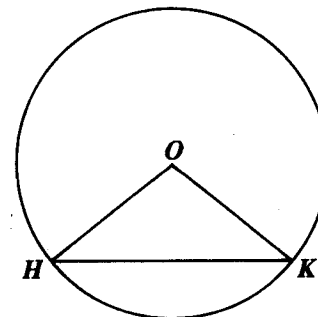
6. The frequency distribution of the Mathematics marks obtained by 100 candidates is given below.

MARKS	NUMBER OF CANDIDATES	CUMULATIVE FREQUENCY
0 – 10	4	4
11 – 20	7	11
21 – 30	9	
31 – 40	12	
41 – 50	18	
51 – 60	13	
61 – 70	12	
71 – 80	11	
81 – 90	9	
91 – 100	5	

- (a) Copy the table above in your answer booklet and complete the cumulative frequency column.
- (b) Using a scale of 1 cm to represent 10 marks on the x -axis and 1 cm to represent 5 candidates on the y -axis, draw the cumulative frequency curve for the data.
- (c) From your cumulative frequency curve, estimate
- the number of candidates who scored at LEAST 45 marks
 - the probability that a candidate chosen at random scored less than 45 marks.
- (11 marks)

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7. In this question, take $\pi = 3.14$



In the figure above, **not drawn to scale**, the chord HK subtends angle HOK at O , the centre of the circle. Angle $HOK = 120^\circ$ and $OH = 12$ cm.

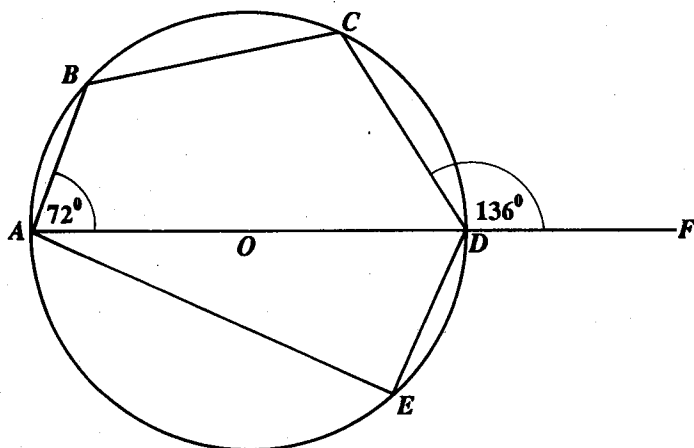
Calculate to three significant figures

- the area of the circle
- the area of the minor sector OHK
- the area of triangle HOK
- the length of the minor arc HK .

(11 marks)

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8.



In the figure above, not drawn to scale, $ABCDE$ is a pentagon inscribed in a circle of centre O . The diameter AD is produced to F . Angle $CDF = 136^\circ$ and angle $BAD = 72^\circ$.

- (a) Calculate, giving reasons for your answer, the magnitude of angles
- CDA
 - BCD
 - AED
- (6 marks)
- (b) Given that $OA = 15$ cm and angle $EAD = 35^\circ$, calculate the length of AE .
- (4 marks)

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SECTION II

Answer TWO questions in this section.

RELATIONS AND FUNCTIONS

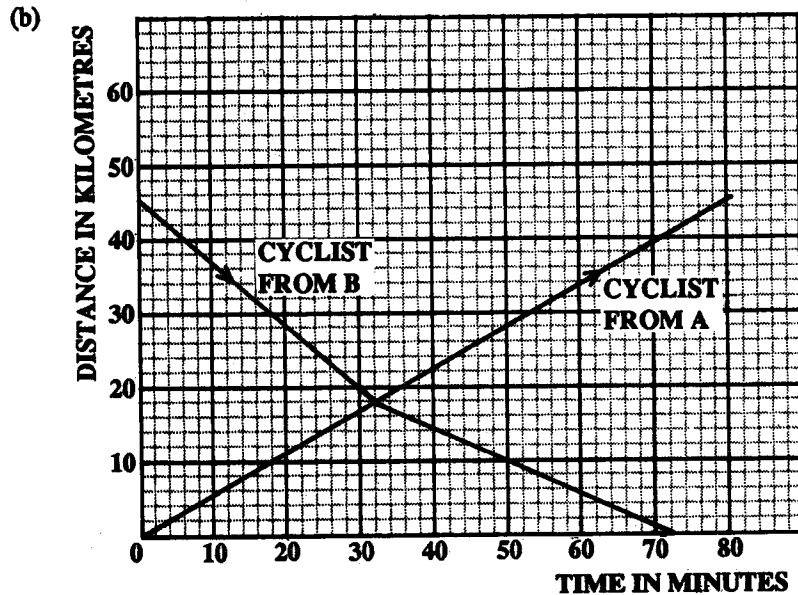
9. A manufacturer produces two types of ball-point pens: Type L and Type M . There are at least 50 of Type L and at least 25 of Type M pens.

The manufacturer, however, does not produce more than 80 of Type L or more than 60 of Type M or more than 120 of both Type L and Type M taken together.

- Using x to represent the number of Type L pens produced and y to represent the number of Type M pens produced, write THREE inequalities (not including $x \geq 0$ and $y \geq 0$) which represent the above conditions.
 - Using a scale of 1 cm to represent 10 pens on EACH axis, draw the graph of the inequalities.
Identify the region which satisfies the inequalities.
 - The manufacturer makes a profit of \$1.50 on each Type L pen and \$1.10 on each Type M pen.
 - Write an expression to represent his total profit.
 - Use the graph to determine the values of x and y which give a maximum profit, and hence determine the maximum profit.
- (15 marks)

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10. (a) Solve the equation $3x^2 + 5x = 6$, giving your answer correct to two decimal places. (5 marks)



The graph above records the journeys of two cyclists travelling between towns A and B. The cyclists begin their journeys at the same time.

Calculate

- the distance between the two towns
 - the time the cyclist from B takes for the journey
 - the average speed of the cyclist from B, in metres per second
 - the distance from town B where the cyclists met
 - the average speed, in metres per second, at which the cyclist from A would need to travel after he met the cyclist from B, in order to complete the journey in the same time as the cyclist from B.
- (10 marks)

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TRIGONOMETRY

11. (a) Prove that $2 \cos^2 \theta - 1 = 1 - 2 \sin^2 \theta$. (2 marks)
- (b) The angles of depression from the top of a tower T to R and S are 32° and 22° respectively. The points R and S and the foot of the tower are on the same horizontal plane. The height of the tower TX is 52 m. The bearings of R and S from X are 270° and 220° respectively.
- Draw a sketch to represent the information given above.
 - Hence or otherwise, calculate
 - the distance RS to one decimal place
 - the bearing of S from R .
- (13 marks)
12. In this question take the radius of the earth to be 6400 km and π to be 3.142.
- (a) The coordinates of the points P and Q on the earth's surface are $(26^\circ \text{ S}, 25^\circ \text{ W})$ and $(60^\circ \text{ N}, 25^\circ \text{ W})$ respectively.
- Calculate
- the shortest distance from P to Q
 - the circumference of the circle of latitude 60° N .
- (7 marks)
- (b) Two tracking stations X and Y are both situated on latitude 60° N . Station X is situated at $(60^\circ \text{ N}, 10^\circ \text{ E})$ and station Y is situated west of X . The distance between X and Y along the latitude 60° N is 1800 km. Calculate the position of the tracking station Y . (8 marks)

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VECTORS AND MATRICES

13. (a) Points U , V and W have position vectors $\begin{pmatrix} -1 \\ -2 \end{pmatrix}$, $\begin{pmatrix} 2 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} 5 \\ \lambda \end{pmatrix}$ respectively. If U , V and W are collinear, determine the value of λ . (6 marks)
- (b) The matrix $\begin{pmatrix} 5 & 12 \\ 3 & 5 \end{pmatrix}$ represents a transformation T .
- (i) Determine the coordinates of the image of the point $(1, -1)$ under the transformation T .
- (ii) Derive the equation of the line onto which the line $x + y = 0$ is mapped by the transformation T . (9 marks)
14. (a) The position vectors of the points A, B, C and D are $\underline{a}$, $\underline{b}$, $3\underline{a} - \underline{b}$ and $\underline{a} + \underline{b}$ respectively.
- (i) Prove that CD is parallel to AB .
- (ii) Determine the ratio $AB : CD$. (8 marks)
- (b) The point $E(3, 2)$, under a shear S , where the invariant line is the x -axis, has image $E'(7, 2)$.
- Determine
- (i) the scale factor of the shear S
- (ii) the (2×2) matrix which represents S
- (iii) the coordinates of the image F' of the point $F(2, 1)$ under the transformation S . (7 marks)

END OF TEST